

Gauge Selection and Reachability in Rational Certificate Search

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Abstract

We study how representation affects rational certificate search in creative telescoping. Experiments across multiple hypergeometric families suggest that gauge selection can change a search problem from unreachable to reachable under fixed ansatz budgets. We further show that recurrence-derived denominator guidance substantially reduces search cost once reachability is established.

1 Introduction

Creative telescoping reduces a hypergeometric summation identity to finding a rational function $R(n, k)$ satisfying

$$F(n+1, k) - F(n, k) = \Delta_k[R(n, k)F(n, k)].$$

Classical results establish algorithmic solvability for broad holonomic classes [9, 10, 6, 1]. However, practical search is bounded by an ansatz: denominator candidates, maximum numerator degree, and recurrence order.

The central empirical observation is representation-dependent reachability:

same identity $\Rightarrow$ different gauge $\Rightarrow$ reachable vs. unreachable under the same budget.

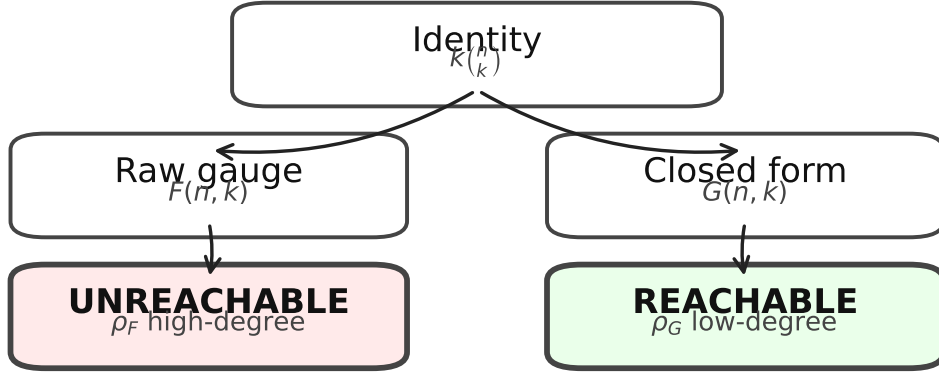
Two phenomena are separated throughout:

- **Phenomenon A (reachability):** whether a certificate is found under a fixed budget.
- **Phenomenon B (cost):** how expensive search is once a reachable gauge is used.
- **Phenomenon C (transfer):** whether structural guidance generalizes across independent families.

2 Related Work

Creative telescoping and WZ methods are foundational in symbolic summation [9, 10, 6]. Extensions to broader holonomic settings and algorithmic implementations are well-developed [1, 3, 7]. D-finite and holonomic function theory provides the structural framework for guaranteed recurrence existence [8, 5]. Standard references in symbolic computation and hypergeometric summation include [2, 4].

Most prior work emphasizes existence, completeness classes, and algorithm design. Our contribution is orthogonal: under bounded ansatz regimes, representation choice (gauge) appears to govern practical reachability, and recurrence-derived denominator guidance appears to reduce cost substantially.



same identity + same budget + different gauge
→ reachable vs unreachable
 ρ_F : shift ratio for $F(n, k)$ ρ_G : shift ratio for $G(n, k)$

Figure 1: The core observation: same identity and same budget, but different gauges lead to qualitatively different reachability outcomes.

3 Background and Definitions

3.1 Families

We study:

- Binomial-power: $F_r(n, k) = \binom{n}{k}^r$, $r \in \{1, 2, 3, 4\}$.
- Weighted-binom: $F(n, k) = k \binom{n}{k}$ with closed-form normalized gauge $G(n, k) = \frac{k \binom{n}{k}}{n 2^{n-1}}$.
- Central Delannoy family (independent order-2 recurrence benchmark).

3.2 Gauge

Given hypergeometric $F(n, k)$, we define a gauge-transformed summand by

$$G(n, k) = c(n, k)F(n, k), \quad c \in \mathbb{Q}(n, k).$$

We compare raw, closed-form, and recurrence-LHS gauges.

3.3 Reachability under budget

Fix budget $B = (D, d, s)$ with denominator set D , max certificate degree d , and max recurrence order s . A certificate is reachable under B iff search finds rational $R(n, k)$ within B and symbolic telescoping verification succeeds.

4 Structural Intuition

Let $\rho_F(n, k) = F(n+1, k)/F(n, k)$. Telescoping coefficients inherit complexity from ρ_F . Under raw combinatorial gauges, high-degree factors often persist; under closed-form normalization,

$$\rho_G = \frac{c(n+1, k)}{c(n, k)} \rho_F,$$

which can cancel dominant factors and lower certificate complexity. This motivates gauge selection as symbolic preconditioning.

5 Phenomenon A: Gauge Sensitivity and Reachability

For $\sum_k k \binom{n}{k} = n2^{n-1}$ (guided denominator ansatz, max degree 4):

Gauge	Reachable	Degree	Equations solved	Median time (s)
raw	NO	—	533	1.530
closed_form	YES	1	10	0.033
recurrence_lhs	NO	—	517	1.920

Table 1: Gauge sensitivity under fixed budget for the weighted-binom identity.

Recovered closed-form certificate:

$$R(n, k) = \frac{k-1}{2(k-n-1)}.$$

This controlled intervention (one representation change, one outcome change) is consistent with a causal interpretation of gauge-dependent reachability.

6 Phenomenon B: Guided Denominator Search Cost

For $\sum_k \binom{n}{k}^r$ with closed-form gauge:

r	Baseline cands	Guided cands	Reduction (%)	Baseline time (s)	Guided time (s)	Time red. (%)	Reach- able?	Verified
1	23	6	74	0.063	0.028	56	YES	YES
2	27	6	78	8.890	1.610	82	YES	YES
3	46	8	83	185.200	33.000	82	NO	NO
4	51	9	82	421.200	77.400	82	NO	NO

Table 2: Guided denominator selection reduces candidate set size and runtime in the binomial-power family.

Delannoy cross-family replication: baseline 22.617 s, guided 2.339 s (89.7% reduction).

7 Discussion

7.1 Qualitative boundary, not only speed

Under bounded search, representation changes can move an instance across a qualitative reachability boundary, not merely accelerate an otherwise-successful run.

7.2 Preconditioning analogy

As numerical preconditioners improve linear-system geometry, gauge changes appear to improve symbolic search geometry by reducing certificate degree and denominator complexity.

7.3 Link between A, B, and C

Phenomenon A concerns feasibility under budget; Phenomenon B concerns efficiency within feasible regimes; Phenomenon C indicates transfer of structural guidance across families.

8 Limitations and Open Questions

- Guided denominator prediction is heuristic; no completeness proof is claimed.
- Higher-order cases ($r = 3, 4$) remain unverified under current budgets.
- Failure under budget B is not non-existence.
- Closed-form gauge selection is currently manual.
- Experiments use univariate recurrences in n (orders 1 and 2 in the present benchmarks); multivariate settings are open.

9 Conclusion

Representation strongly influences certificate reachability in the tested regime. Once representation enables reachability, structural guidance promotes efficiency. This yields a practical workflow for bounded symbolic search: first optimize representation, then optimize denominator search.

A Reproducibility

Experiments are generated by `src/wz_lab.py`, with JSON/CSV traces in `results/`. Negative results are retained with full metadata (gauge, denominator candidates, equations solved, runtime), enabling explicit reachability-boundary analysis.

Code, traces, and experiment scripts are available at: github.com/JoseClaudioSJr/gauge-reachability-symbolic-search

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